

1. Convert the following decimals into fractions:

(i) 0.75

- $0.75 = \frac{75}{100} = \frac{3}{4}$

(ii) 0.625

- $0.625 = \frac{625}{1000} = \frac{5}{8}$

(iii) 2.35

- $2.35 = 2 + \frac{35}{100} = 2 + \frac{7}{20} = \frac{47}{10}$

(iv) 0.333

- $0.333 \approx \frac{333}{1000}$. $0.333 \approx \frac{1}{3}$ (This is an approximation; $\frac{1}{3}$ is exact.)

(v) 0.85

- $0.85 = \frac{85}{100} = \frac{17}{20}$

(vi) 0.475

- $0.475 = \frac{475}{1000} = \frac{19}{200}$
-

2. Convert the following fractions into decimals:

(i) $\frac{3}{4}$

- $\frac{3}{4} = 0.75$

(ii) $\frac{8}{5}$

- $\frac{8}{5} = 1.6$

(iii) $\frac{16}{20}$

- $\frac{16}{20} = \frac{4}{5} = 0.8$

(iv) $\frac{3}{5}$

- $\frac{3}{5} = 0.6$

(v) $\frac{1}{2}$

- $\frac{1}{2} = 0.5$

(vi) $\frac{7}{10}$

- $710=0.7107=0.7$

3. Solve the following expressions using the BODMAS rule:

(i) $(3+8) \times 4 - (6+2 \times 5) - 7 \times 2 - 1$

1. Parentheses:

- $3+8=11$
- $2 \times 5=10$, then $6+10=16$

2. Multiplication:

- $11 \times 4=44$
- $7 \times 2=14$

3. Subtraction:

- $44-16=28$
- $28-14=14$
- $14-1=13$

Answer: 13

(ii) $(2+7) \times 5 - (-3+1)$

1. Parentheses:

- $2+7=9$
- $-3+1=-2$

2. Multiplication:

- $9 \times 5=45$

3. Subtraction:

- $45 - (-2) = 45 + 2 = 47$

Answer: 47

(iii) $-3 \times (4-2) \times (-1-7) \times (-2+3) \times 8$

1. Parentheses:

- $4-2=2$

- $-1-7=-8-1-7=-8$

- $-2+3=1-2+3=1$

2. Multiplication (left to right):

- $-3 \times 2 = -6 - 3 \times 2 = -6$

- $-6 \times (-8) = 48 - 6 \times (-8) = 48$

- $48 \times 1 = 48 48 \times 1 = 48$

- $48 \times 8 = 384 48 \times 8 = 384$

Answer: 384384

(iv) $-3 \times (4-2) \times (-1-7) \times (-2+3) + 8 - 3 \times (4-2) \times (-1-7) \times (-2+3) + 8$

1. Parentheses (Brackets):

- $4-2=2 4-2=2$

- $-1-7=-8-1-7=-8$

- $-2+3=1-2+3=1$

2. Multiplication:

- $-3 \times 2 = -6 - 3 \times 2 = -6$

- $-6 \times (-8) = 48 - 6 \times (-8) = 48$

- $48 \times 1 = 48 48 \times 1 = 48$

3. Addition:

- $48+8=56 48+8=56$

Answer: 5656

(v) $4+9 \times (6+2) \cdot 52 + 9 \times (6+2) \cdot 52$

1. Parentheses:

- $6+2=8 6+2=8$

2. Orders (Exponents):

- $52=25 52=25$

3. Multiplication:

- $9 \times 8 = 72$
- $72 \times 25 = 1800$

4. **Addition:**

- $4 + 1800 = 1804$
- Answer: 1804**

(vi) $(5+2) \times (8+4) - 23$

1. **Parentheses:**

- $5 + 2 = 7$
- $8 + 4 = 12$

2. **Orders:**

- $23 = 23$

3. **Multiplication:**

- $7 \times 12 = 84$

4. **Subtraction:**

- $84 - 8 = 76$
- Answer: 76**

(vii) $32 \times 23 + 52 \times 52 + 2850$

1. **Orders:**

- $32 = 32$
- $23 = 23$
- $52 = 52$
- $2850 = 2850$ (Any number to the power of 0 is 1.)

2. **Multiplication:**

- $9 \times 8 = 72$
- $25 \times 25 = 625$

3. Addition:

- $72+625=697$
- $697+1=698$

Answer: 698

(viii) $36+[12+8-9(4\times 2-3+(2\times 8+4+4))]$

1. Innermost Parentheses:

- $2\times 8=16$
- $16+4+4=24$

2. Next Parentheses:

- $4\times 2=8$
- $8-3+24=29$

3. Multiplication:

- $9\times 29=261$

4. Addition/Subtraction Inside Brackets:

- $12+8=20$
- $20-261=-241$

5. Final Addition:

- $36+(-241)=-205$
- Answer: -205**
-

(ix) $8+[3+8-9(8\times 2-3+(2\times 4+2+2))]$

1. Innermost Parentheses:

- $2\times 4=8$
- $8+2+2=12$

2. Next Parentheses:

- $8\times 2=16$

- $16-3+12=25$ $16-3+12=25$

3. Multiplication:

- $9 \times 25 = 225$ $9 \times 25 = 225$

4. Addition/Subtraction Inside Brackets:

- $3+8=11$ $3+8=11$

- $11-225=-214$ $11-225=-214$

5. Final Addition:

- $8+(-214)=-206$ $8+(-214)=-206$

Answer: -206

(x) $\{10+16+(14+7+2)\}+\{(5 \times 22+2) \times 5\}$ $\{10+16+(14+7+2)\}+\{(5 \times 22+2) \times 5\}$

1. First Curly Braces:

- $14+7+2=23$ $14+7+2=23$

- $10+16+23=49$ $10+16+23=49$

2. Second Curly Braces:

- **Orders:**

- $22=42$ $22=4$

- $25=52$ $25=5$

- **Multiplication/Division:**

- $5 \times 4 = 20$ $5 \times 4 = 20$

- $20+2=22$ $20+2=22$

- $55=15$ $55=1$

- $22 \times 1 = 22$ $22 \times 1 = 22$

3. Final Addition:

$49+22=71$ $49+22=71$

Answer: 71

4. Birthday Party Budget

- **Sandwiches:**
 $30 \text{ persons} \times 2 \text{ sandwiches/person} = 60 \text{ sandwiches}$
 $60 \times \text{Rs } 500 = \text{Rs } 30,000$
 - **Coffee:**
 $30 \times \text{Rs } 200 = \text{Rs } 6,000$
 - **Total Cost:**
 $\text{Rs } 30,000 + \text{Rs } 6,000 = \text{Rs } 36,000$
-

5. Vacation Trip Budget

- **Airfare:** Rs 400,000
 - **Accommodation:**
 $5 \text{ nights} \times \text{Rs } 20,000 = \text{Rs } 100,000$
 - **Food:**
 $5 \text{ days} \times \text{Rs } 8,500 = \text{Rs } 42,500$
 - **Total Budget:**
 $\text{Rs } 400,000 + \text{Rs } 100,000 + \text{Rs } 42,500 = \text{Rs } 542,500$
-

6. Hypermarket Purchase

- **Apples:**
 $5 \times \text{Rs } 50 = \text{Rs } 250$
 - **Oranges:**
 $10 \times \text{Rs } 40 = \text{Rs } 400$
 - **Bread:**
 $2 \times \text{Rs } 200 = \text{Rs } 400$
 - **Total Cost:**
 $\text{Rs } 250 + \text{Rs } 400 + \text{Rs } 400 = \text{Rs } 1,050$
-

7. Restaurant Bill with Tip

- **Bill:** Rs 8,000
 - **Tip (5%):**
 $0.05 \times \text{Rs } 8,000 = \text{Rs } 400$ $0.05 \times \text{Rs } 8,000 = \text{Rs } 400$
 - **Total Cost:**
 $\text{Rs } 8,000 + \text{Rs } 400 = \text{Rs } 8,400$ $\text{Rs } 8,000 + \text{Rs } 400 = \text{Rs } 8,400$
-

8. Cake Baking Rate

- **Cakes per Hour:**
 $48 \text{ cakes} / 12 \text{ hours} = 4 \text{ cakes/hour}$ $48 \text{ cakes} / 12 \text{ hours} = 4 \text{ cakes/hour}$
-

9. Car Mileage

- **Miles per Gallon:**
 $500 \text{ miles} / 50 \text{ gallons} = 10 \text{ miles/gallon}$ $500 \text{ miles} / 50 \text{ gallons} = 10 \text{ miles/gallon}$
-

10. Car Speed

- **Speed (mph):**
 $360 \text{ miles} / 6 \text{ hours} = 60 \text{ mph}$ $360 \text{ miles} / 6 \text{ hours} = 60 \text{ mph}$
-

11. Reading Rate

- **Pages per Hour:**
 $60 \text{ pages} / 2 \text{ hours} = 30 \text{ pages/hour}$ $60 \text{ pages} / 2 \text{ hours} = 30 \text{ pages/hour}$
-

12. Runner's Speed

- **Speed (m/s):**
 $200 \text{ meters} / 25 \text{ seconds} = 8 \text{ m/s}$ $200 \text{ meters} / 25 \text{ seconds} = 8 \text{ m/s}$
-

13. Test Points per Question

- **Points per Question:**

100 points 25 questions = 4 points/question 25 questions 100 points = 4 points/question

14. Time to Read 1 Book

- **Time per Book:**

1 week 3 books = 13 week/book 3 books 1 week = 31 week/book
(or **2.33 days/book**)

15. Ratio with Terms Differing by 40

Let the ratio terms be $2x$ and $7x$.

- **Difference:**

$$7x - 2x = 40 \quad 7x - 2x = 40$$

$$5x = 40 \Rightarrow x = 8 \quad 5x = 40 \Rightarrow x = 8$$

- **Ratio:**

$$2x : 7x = 16 : 56 \quad 2x : 7x = 16 : 56$$

Answer: 16:56

16. Total Students in College

- **Given Ratio (Boys:Girls):** 8:5

- **Number of Girls:** 160

- **Let x be the multiplier:**

$$5x = 160 \Rightarrow x = 32 \quad 5x = 160 \Rightarrow x = 32$$

- **Number of Boys:**

$$8x = 8 \times 32 = 256 \quad 8x = 8 \times 32 = 256$$

- **Total Students:**

$$256 + 160 = 416 \quad 256 + 160 = 416$$

17. Profit Sharing Ratio (A and B)

- **Initial Investments:**

- A: Rs 8,000
- B: Rs 9,000
- **After 4 Months:**
 - A withdraws half: Rs 4,000 remaining
 - B continues: Rs 9,000
- **After 6 Months (2 months later):**
 - B withdraws 1331: Rs 6,000 remaining
- **Effective Investments:**
 - A: $(8,000 \times 4) + (4,000 \times 8) = 32,000 + 32,000 = 64,000$
 - B: $(9,000 \times 6) + (6,000 \times 6) = 54,000 + 36,000 = 90,000$

Profit Ratio:

$$64,000:90,000 = 32:45$$

1. Class of 39 Students (Ratio of Boys and Girls)

- **Given:**
 - Total students = 39
 - Ratio of boys to girls = 4:4 (simplifies to **1:1**)
- **Number of Boys and Girls:**

$$39 \div 2 = 19.5$$

→ Not possible (must be whole numbers).

Correction: The ratio is likely **4:5** (total parts = 9).

 - Boys = $4 \times 39 \div 9 = 17.33$ → Invalid.

Recheck the ratio or total students.

I. Ratio of Boys and Girls in a School (9:5)

- **Total students = 1050**
- **Boys = $9 \times 1050 \div 14 = 675$**

I. Pens and Pencils in a Shop (Ratio 3:2)

- **Average of Pens and Pencils = 180**
$$\frac{\text{Pens} + \text{Pencils}}{2} = 180 \quad \text{Pens} + \text{Pencils} = 360$$
- **Let Pens = $3x$, Pencils = $2x$**
$$3x + 2x = 360 \quad 5x = 360 \quad x = 72$$
- **Pencils = $2x = 144$**

I. Social Work Group (64 Boys and 40 Girls)

- **Let new members = x (same for boys and girls)**
- **New ratio (Boys:Girls) = 43:7**
$$\frac{64+x}{40+x} = \frac{43}{7} \quad 7(64+x) = 43(40+x)$$

Cross-multiply:
$$448 + 7x = 1720 + 43x \quad -36x = 1272 \quad x = -35.33$$

 $x = -35.33 \rightarrow$ **Invalid** (Members cannot be negative).
Possible error in ratio or initial numbers.

2. Ratio of Ages of Two Boys (3:4)

- **Let ages = $3x$ and $4x$**
- **After 3 years:**
$$\frac{3x+3}{4x+3} = \frac{5}{4} \quad 4(3x+3) = 5(4x+3)$$

Cross-multiply:
$$12x + 12 = 20x + 15 \quad -8x = 3 \quad x = -0.375$$
- **Ages after 21 years:**
$$\frac{3x+21}{4x+21} = \frac{10}{11} \quad 11(3x+21) = 10(4x+21)$$

$$33x + 231 = 40x + 210 \quad -7x = -21 \quad x = 3$$

Ratio = 10:11

3. Money Divided Between A and B (Ratio 5:6)

- **B gets Rs 360:**
 $611 \times \text{Total} = 360$
 $116 \times \text{Total} = 360$
 $\text{Total} = 360 \times 116 = 660$
 $\text{Total} = 360 \times 611 = 660$

4. Rectangle with Golden Ratio (Length-to-Width)

- **Golden Ratio (ϕ) ≈ 1.618**
- **Given width = w , length = $1.618w$**

5. Rectangle with Length-to-Width = ϕ

- **Width = w units, Length = ϕw**

6. Rectangle (10cm $\times$ 16cm) Adjusted to Golden Ratio

- **Current ratio (longer:shorter) = $16:10 = 1.6$**
- **Golden ratio = $1.618 \rightarrow$ Already close.**
Adjust shorter side (s):
 $16s = 1.618s$
 $16 = 1.618$
 $s = 16 \div 1.618 \approx 9.89 \text{ cm}$

7. Golden Ratio Approximations via Fibonacci Sequence

- **Fibonacci sequence:** 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144...
- **Approximations (F_{n+1}/F_n):**
 1. $2/1 = 2$
 2. $3/2 = 1.5$
 3. $5/3 \approx 1.666$
 4. $8/5 = 1.6$

5. $138 \approx 1.625813 \approx 1.625$
Converges to $\phi \approx 1.618$
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8. Clock with Golden Ratio Minute Hand

- Radius = 12 cm
 - Minute hand length = $\phi \times 12 \approx 19.42$ cm
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9. Photo Frame with Golden Ratio Height

- Width = 24 inches
 - Height = $\phi \times 24 \approx 38.83$ inches
-

10. Spiral from Golden Rectangles (10th Side)

- Initial sides: 1, 1.618
 - Fibonacci growth:
 $F_{10} = 55$ (using Fibonacci sequence)
 10th side ≈ 55 units
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11. Circular Garden Path with Golden Ratio

- Garden diameter = 20 ft $\rightarrow$ Radius = 10 ft
 - Path width (w) to diameter ratio = ϕ :
 $w = \phi \times 10 \approx 16.18$ ft
 Path radius = Garden radius + w = 26.18 ft
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12. Proportions:

(i) $X:7::12:21$

$$X \cdot 21 = 7 \cdot 12 \Rightarrow X = \frac{7 \cdot 12}{21} = 4$$

(ii) $4:X::8:14$

$$4X=814 \Rightarrow X=4 \times 148=568=7X4=148 \Rightarrow X=84 \times 14=856=7$$

(iii) 6:9::X:156:9::X:15

$$69=X15 \Rightarrow X=6 \times 159=909=1096=15X \Rightarrow X=96 \times 15=990=10$$

(iv) 12:18::20:X12:18::20:X

$$1218=20X \Rightarrow X=18 \times 2012=36012=301812=X20 \Rightarrow X=1218 \times 20=12360=30$$

(v) X:1.5::6.3:4.5X:1.5::6.3:4.5

$$X1.5=6.34.5 \Rightarrow X=1.5 \times 6.34.5=9.454.5=2.11.5X=4.56.3 \Rightarrow X=4.51.5 \times 6.3=4.59.45=2.1$$

(vi) 6:12::X:246:12::X:24

$$612=X24 \Rightarrow X=6 \times 2412=14412=12126=24X \Rightarrow X=126 \times 24=12144=12$$

(vii) 25:X::5:625:X::5:6

$$25X=56 \Rightarrow X=25 \times 65=1505=30X25=65 \Rightarrow X=525 \times 6=5150=30$$

(viii) 30:5::42:X30:5::42:X

$$305=42X \Rightarrow X=5 \times 4230=21030=7530=X42 \Rightarrow X=305 \times 42=30210=7$$

33. Proportional Calculations

(i) Third Proportional to 9 and 12

- **Concept:** If $a:b::b:c$, then c is the third proportional.
- **Given:** $9:12::12:c$
- **Calculation:**

$$912=12c \Rightarrow c=12 \times 129=1449=16129=c12 \Rightarrow c=912 \times 12=9144=16$$

(ii) Third Proportional to 16 and 36

- **Given:** $16:36::36:c$
- **Calculation:**

$$1636=36c \Rightarrow c=36 \times 3616=129616=813616=c36 \Rightarrow c=1636 \times 36=161296=81$$

(iii) Mean Proportional Between 6 and 24

- **Concept:** The mean proportional bb satisfies $a:b::b:ca:b::b:c$.
- **Given:** $6:b::b:24$
- **Calculation:**

$$b^2 = 6 \times 24 = 144 \Rightarrow b = \sqrt{144} = 12$$

(iv) Mean Proportional Between 7 and 28

- **Given:** $7:b::b:28$
- **Calculation:**

$$b^2 = 7 \times 28 = 196 \Rightarrow b = \sqrt{196} = 14$$

34. Combined Ratios

(i) Given $a:b=4:3$ and $b:c=5:3$

- **Step 1:** Make the common term b equal in both ratios.
 - $a:b=4:3$ (Multiply by 5) $\rightarrow 20:15$
 - $b:c=5:3$ (Multiply by 3) $\rightarrow 15:9$
- **Combined Ratio:** $a:b:c=20:15:9$
- **Answer:** $20:15:9$

(ii) Given $A:B=2:3$ and $B:C=4:5$

- **Step 1:** Make the common term B equal in both ratios.
 - $A:B=2:3$ (Multiply by 4) $\rightarrow 8:12$
 - $B:C=4:5$ (Multiply by 3) $\rightarrow 12:15$
- **Combined Ratio:** $A:B:C=8:12:15$
- **Answer:** $8:12:15$

35. Solving Proportions

(i) Given $A:B=4:5$ and $B:C=6:7$

- **Step 1:** Make the common term B equal in both ratios.

- $A:B=4:5$ $A:B=4:5$ (Multiply by 6) $\rightarrow 24:30$
- $B:C=6:7$ $B:C=6:7$ (Multiply by 5) $\rightarrow 30:35$
- **Combined Ratio:** $A:B:C=24:30:35$
- **Answer:** $A:C=24:35$

(ii) Given $A:B=32:35$ and $B:C=21:32$

- **Step 1:** Make the common term B equal in both ratios.
 - $A:B=32:35$ (Multiply by 21) $\rightarrow 672:735$
 - $B:C=21:32$ (Multiply by 35) $\rightarrow 735:1120$
- **Combined Ratio:** $A:B:C=672:735:1120$
- **Simplify:** Divide by 224 $\rightarrow 3:5:5$
- **Answer:** $A:C=3:5$

(iii) Given $3A=2B=4C$

- Let $3A=2B=4C=k$
 - $A=k/3$, $B=k/2$, $C=k/4$
- Find LCM of denominators (3, 2, 4): 12
- Multiply each term by 12:
 - $A=4$, $B=6$, $C=3$
- **Ratio:** $A:B:C=4:6:3$
- **Answer:** $4:6:3$

(iv) Given $6A=4B=9C$

- Let $6A=4B=9C=k$
 - $A=k/6$, $B=k/4$, $C=k/9$
- Find LCM of denominators (6, 4, 9): 36
- Multiply each term by 36:
 - $A=6$, $B=9$, $C=4$
- **Ratio:** $A:B:C=6:9:4$

- **Answer:** 6:9:46:9:4

(v) Given $A=2B=4C$

- **Let $A=2B=4C=k$**
 - $A=k, B=2k, C=4k$
- **Find LCM of denominators (1, 2, 4): 4**
- **Multiply each term by 4:**
 - $A=4k, B=2k, C=k$
- **Ratio: $A:B:C=4:2:1$**
- **Answer: 4:2:1**

Summary of Answers:

Question	Answer
33 (i)	16
33 (ii)	81
33 (iii)	12
33 (iv)	14
34 (i)	20:15:9
34 (ii)	8:12:15
35 (i)	24:35
35 (ii)	3:5
35 (iii)	4:6:3

Question	Answer
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35 (iv)	6:9:4
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35 (v)	4:2:1
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36. Flower Garden Length

- **Given:**
 - Ratio of length to width = 3:2
 - Width = 36 m
- **Let length = $3x$, width = $2x$.**
 $2x=36 \Rightarrow x=18$
 $3x=36 \Rightarrow x=18$
- **Length = $3 \times 18 = 54$ m**
 $3 \times 18 = 54$ m.

37. Bag of Coins

- **Given:**
 - Ratio of 50 P, 25 P, 10 P coins = 5:9:4
 - Total amount = Rs. 206
 - **Let number of coins be $5x$, $9x$, $4x$.**
 - **Total value:**
 $0.50 \times 5x + 0.25 \times 9x + 0.10 \times 4x = 206$
 $0.50 \times 5x + 0.25 \times 9x + 0.10 \times 4x = 206$
 $2.5x + 2.25x + 0.4x = 206$
 $2.5x + 2.25x + 0.4x = 206$
 $5.15x = 206 \Rightarrow x = 40$
 $5.15x = 206 \Rightarrow x = 40$
 - **Number of coins:**
 - 50 P: $5 \times 40 = 200$
 $5 \times 40 = 200$
 - 25 P: $9 \times 40 = 360$
 $9 \times 40 = 360$
 - 10 P: $4 \times 40 = 160$
 $4 \times 40 = 160$
-

38. Salaries of Mr. A and Mr. B

- **Given:**
 - Original ratio (A:B) = 2:3
 - After increase of Rs. 4000 each, new ratio = 40:57
 - **Let original salaries be $2x$ and $3x$.**
 - **New salaries:**
 $2x + 4000$ and $3x + 4000$
 $40(2x + 4000) = 57(3x + 4000)$
Cross-multiply:
 $40(2x + 4000) = 57(3x + 4000)$
 $80x + 160000 = 171x + 228000$
 $-91x = -68000 \Rightarrow x = \frac{68000}{91} \approx 747.25$
 - **B's salary = $3x = 3 \times \frac{68000}{91} = \frac{204000}{91} \approx 2241.76$ Rs.**
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39. Division of Rs. 782 in Ratio 12:23:34

- **Simplify the ratio:**
Multiply by 12 (LCM of denominators):
6:8:9
 - **Total parts = $6 + 8 + 9 = 23$**
 - **First part = $\frac{6}{23} \times 782 = 204$ Rs**
-

40. Mango Juice and Water Mixture

- **Given:**
 - Original ratio (Mango:Water) = 4:3
 - After adding 5 L water, new ratio = 4:5
- **Let original quantities be $4x$ and $3x$.**
 - **New water quantity:** $3x + 5$
 - **New ratio:**
 $4x : 3x + 5 = 4 : 5$
 $4(3x + 5) = 5(4x)$
 $12x + 20 = 20x$
 $-8x = -20 \Rightarrow x = \frac{20}{8} = 2.5$

Cross-multiply:

$$20x = 12x + 20 \Rightarrow 8x = 20 \Rightarrow x = 2.5$$
$$20x = 12x + 20 \Rightarrow 8x = 20 \Rightarrow x = 2.5$$

- **Mango juice** = $4x = 4 \times 2.5 = 10$ L $4x = 4 \times 2.5 = 10$ L.
-

41. Ages of Friends A, B, and C

- **Given:**
 - Ratio (A:B) = 3:5
 - Ratio (B:C) = 3:5
 - Total age = 147
 - **Combine ratios:**
 $A:B:C = 9:15:25$ $A:B:C = 9:15:25$
 - **Total parts** = $9+15+25=49$ $9+15+25=49$
 - **B's age** = $15/49 \times 147 = 45$ years $15/49 \times 147 = 45$ years.
-

42. Exercise Books in Shop

- **Given:**
 - Ratio (Pencils:Pens:Exercise books) = 10:2:3
 - Pencils = 120
 - **Let parts be $10x$, $2x$, $3x$.**
 $10x = 120 \Rightarrow x = 12$ $10x = 120 \Rightarrow x = 12$
 - **Exercise books** = $3 \times 12 = 36$ $3 \times 12 = 36$.
-

43. Incomes of A and B

- **Given:**
 - Income ratio (A:B) = 3:4
 - Each saves Rs. 100/month
 - Expenditure ratio = 1:2

$$y^2 81 / y = 243 \Rightarrow y^3 = 19683 \Rightarrow y = 27 \quad 81 / y^2 = 243 \Rightarrow y^3 = 19683 \Rightarrow y = 27$$

$$x = 81 \quad 27 = 3x = 27 \quad 81 = 3$$

- **Values:**

$$x = 3 \quad x = 3, \quad y = 27 \quad y = 27.$$

46. Numbers with Mean Proportion 8 and Third Proportion 64

- **Mean proportion:**

$$xy = 8 \Rightarrow xy = 64 \quad xy = 8 \Rightarrow xy = 64$$

- **Third proportion:**

$$y^2 x = 64 \quad xy^2 = 64$$

- **Solve:**

$$\text{From } xy = 64 \quad xy = 64, \quad x = 64 / y \quad yx = y^64.$$

Substitute:

$$y^2 64 / y = 64 \Rightarrow y^3 = 4096 \Rightarrow y = 16 \quad 64 / y^2 = 64 \Rightarrow y^3 = 4096 \Rightarrow y = 16$$

$$x = 64 / 16 = 4 \quad x = 16 / 4 = 4$$

- **Values:**

$$x = 4 \quad x = 4, \quad y = 16 \quad y = 16.$$

47. 60% of 800

- **Calculation:**

$$0.60 \times 800 = 480 \quad 0.60 \times 800 = 480.$$